Stanley Nelavala

Software Engineer | Python, FastAPI, React, AWS
Huntington, IN | 218-213-8220 | consultantstanleymn@gmail.com | linkedin.com/in/stanley-sujith-nelavala | stanley-n.com | github.com/consultantstanleymn

SUMMARY

Software Development Engineer with one year at a 10-person hospital operations startup, building Python and FastAPI APIs, React web apps and a React Native mobile app. Cut a deep catalog query from 285 ms to 0.20 ms with keyset paging, fixed a regex denial-of-service bug, and raised mobile test coverage from 14% to 93%. Led the service-to-service API integration across every module, RxDesign, and the RxAsset mobile app. M.S. in Computer Science (4.0 GPA) and AWS Certified Solutions Architect, Associate.

EXPERIENCE

Software Development Engineer, Mass Technologies (startup), Huntington, IN (in person). Nov 2025 to Present
Team of 10 building RxApps360, a multi-tenant platform for hospital operations. Not yet launched, so all measurements below come from development and test environments. Proprietary code.

Device catalog and planning APIs (Python, FastAPI, PostgreSQL, Redis)

- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms.
- Built the catalog search API on a shared service layer with keyset cursors and a unique-id tie-break, so pages never skip or repeat rows; tested with rows inserted, deleted and tied between pages.
- Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search and cursor paging.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.

Asset tracking mobile app (React Native, Expo, FastAPI)

- Built the RxAsset mobile app for iOS and Android.
- Resolved 27 defects across the asset transfer and parts pages.
- Set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission, and raised statement coverage from 14% to 93% (582 tests in 68 suites).
- Built scan-to-search and an on-device voice assistant (speech-to-text and text-to-speech run on the phone, with no cloud call); fixed a native crash in the speech library bridge.

Hospital design service (FastAPI, React, TypeScript)

- Built RxDesign, the hospital design service, backend and web.
- Wrote a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; added 220+ offline tests to the design API.

Platform (AWS, GitHub Actions)

- Load-tested the sign-in flow with JMeter at 100 concurrent users.
- Standardized malware scanning of file uploads (AWS GuardDuty) across 9 app modules.
- Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS), so they run about 53 hours a week instead of 168.

Graduate Research Assistant and Teaching Assistant, University of Minnesota Duluth. Aug 2023 to Aug 2025

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session.
- Led labs and office hours for Python, NLP and Parallel Programming.

Adjunct Faculty (part-time, 8 hours a week), The College of St. Scholastica. Jan 2026 to Aug 2026. Taught Programming II (Java) and Machine Learning for Data Science.

SELECTED PROJECTS

Agent Integration Platform | Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, OpenTelemetry, Terraform | github.com/consultantstanleymn/agent-integration-platform

- Built a LangGraph case workflow with typed state, a Postgres checkpointer, human approval with resume after restart, step and token budgets and a written authority boundary; one constrained MCP write tool; 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction and an append-only audit log.
- Hybrid retrieval (pgvector and Postgres full-text search fused by RRF) with access labels, a 48-query golden set (labels pending review), /v2/search beside /v1, idempotent writes, a Qdrant comparison in an ADR, a Redis rate limiter and embedding cache.
- Failover between two LLM provider adapters with timeouts, retry budget, circuit breaker and token cost records; S3, SQS (DLQ, redrive) and Secrets Manager adapters with emulated integration tests; Terraform modules and teardown scripts; Jaeger traces, a Locust load test, a release manifest with a CycloneDX SBOM, GitHub Actions security and test workflows.

Customer Integration Console | TypeScript, Next.js, Fastify, PostgreSQL, OIDC, Keycloak realm config, Trivy, Docker Compose | github.com/consultantstanleymn/customer-integration-console

- Wrote a STRIDE threat model; generated an authorization matrix test over every route and role; added 10 negative OIDC token tests (expired, wrong audience, unknown key id, alg none, HS256 key confusion, replay); OIDC discovery and client-credentials scopes against a mock IdP.
- Envelope encryption (AES-256-GCM) of source credentials with key rotation and a grace window; hash-chained audit log with export and tamper detection; CSP, security headers and a login rate limit.
- Security gate with gitleaks, npm audit, Trivy image scans, a CycloneDX SBOM and an exceptions file with expiry; 256 tests plus 7 browser tests; GitHub Actions security and test workflows.

Payments Ledger Service | Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API (Redpanda), Prometheus, Grafana, Toxiproxy, k6 | github.com/consultantstanleymn/payments-ledger-service

- Defined three user-facing SLIs with 28-day error budgets; multi-window burn-rate alerts tested with promtool; Grafana SLO dashboard on Prometheus.
- Ran a scripted failure lab (Kafka down, Postgres restart, relay crash, poison event, latency injection with Toxiproxy) with balance invariants checked after each; added a dead-letter topic with once-only redrive.
- Rolled forward and back by image digest across an expand-and-contract Flyway migration; k6 SLO thresholds as a gate; append-only caller audit log; STRIDE review of the money path with a written risk decision.

Inference Gateway | Go, gRPC, ONNX Runtime, OpenTelemetry, Jaeger, Prometheus, Grafana, Locust, k6, Helm | github.com/consultantstanleymn/inference-gateway

- Built a diagnostics stack (OpenTelemetry, Jaeger, Prometheus, Grafana) with a golden-signals dashboard, a cardinality budget with a bounded-labels test, latency exemplars linked to traces, and a Locust profile beside k6.
- Attributed the latency bottleneck with one-factor experiments and pprof, ruled out GC and tokenizer cost, and wrote a capacity model (Little’s law) that sizes a 10x launch peak at 2 to 3 replicas (local benchmark); corrected an unreviewed capacity draft line by line.
- Wrote a canary controller with a weighted proxy and an error and p95 gate that rolled a faulty version back on its own; latency SLO alerts tested with promtool; release manifest with model and runtime hashes and a CycloneDX SBOM.

Webhook and Connector Gateway | Rust, axum, tokio, SQLite, PostgreSQL, sqlx, OpenTelemetry, k6, AES-GCM | github.com/consultantstanleymn/webhook-connector-gateway

- Added a Postgres store (FOR UPDATE SKIP LOCKED claim) beside SQLite behind one store port, with versioned migrations; the full suite of 88 tests runs on both backends.
- Global backpressure (503 with Retry-After), a per-tenant retry budget, and a kill -9 failure lab with 500 of 500 events applied once at the downstream after redrive; delivery SLO alerts tested with promtool; k6 load test and OpenTelemetry spans in Jaeger.
- Encrypted tenant signing secrets at rest (AES-256-GCM) with keys from a secret provider and a reseal command for key rotation.

FHIR Analytics Pipeline | Python, Airflow, dbt, DuckDB, Azure Blob SDK on Azurite, Bicep, HL7 FHIR R4 | github.com/consultantstanleymn/fhir-analytics-pipeline

- Enforced dbt model contracts and source freshness, added a quarantine-rate gate, and published a provenance manifest (input hashes, code SHA, dbt manifest hash, row counts) with every run.
- Added a k-anonymity check (k=5) with generalization and suppression before publishing, versioned HMAC pseudonymization keys with a rotation test, and a HIPAA-style control mapping with evidence links.
- Pipeline metrics with a 06:00 freshness alert tested with promtool; a failure lab for corrupt input, schema drift, late updates and storage outage; a late-update fix proven by dbt tests; 108 tests.

EDUCATION

- M.S. Computer Science, University of Minnesota Duluth, 2023 to 2025. GPA 4.0. Graduate Summer Research Fellowship, 2024 and 2025.
- B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 to 2023.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, Java, Go, Rust
- **Backend and data:** FastAPI, Spring Boot, Fastify, axum, REST APIs and versioning, gRPC, PostgreSQL (pgvector, full-text search, SKIP LOCKED queues), Redis, Kafka API (Redpanda), SQLite, Node.js, MySQL, dbt, Airflow, DuckDB, idempotency, ports and adapters, migrations (Alembic, Flyway, sqlx)
- **AI and agents:** LangGraph, MCP, human-in-the-loop, hybrid retrieval and RAG, deterministic LLM evals, hosted-LLM adapters with failover and cost records, output guardrails, PII redaction, ONNX Runtime
- **Frontend and mobile:** React, Next.js, React Native, Expo
- **Cloud and DevOps:** AWS (ECS, RDS, S3, SQS, Secrets Manager, GuardDuty), Terraform modules, Linux, Docker and Compose, Helm, Kubernetes (kind), Git, GitHub Actions (CI/CD, security workflows)
- **Observability and reliability:** OpenTelemetry, Jaeger, Prometheus, Grafana, SLOs and burn-rate alerts (promtool), capacity planning (Little’s law), pprof, canary rollout and rollback, failure labs and runbooks, Toxiproxy
- **Security:** STRIDE threat modeling, OIDC and OAuth2 (client-credentials scopes, mock IdP, Keycloak realm config), RBAC, envelope encryption (AES-GCM), key rotation, hash-chained audit log, gitleaks, Trivy, CycloneDX SBOM, CodeQL workflows, release manifests, HIPAA-style control mapping, k-anonymity
- **Data engineering:** HL7 FHIR R4, dbt contracts, data provenance, data-quality gates, HMAC pseudonymization, Azure Blob SDK on Azurite, Bicep
- **Testing and practices:** pytest, Jest, Playwright, k6, Locust, jqwik property tests, code review, Agile
- **Machine learning:** BERTopic, PaddleOCR, PyTorch, int8 quantization

CERTIFICATION

AWS Certified Solutions Architect, Associate. Feb 2025 to Feb 2028. Verifiable on Credly.